2025
Higher School Certificate
Trial Examination

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A data sheet, formulae sheets and Periodic Table are provided
- Write your student number and/or name on every page

Total marks – 100

Section I (Pages 2–10)
20 marks
- Attempt Questions 1–20
- Allow about 35 minutes for this part

Section II (Pages 11–31)
80 marks
- Attempt Questions 21–34
- Allow about 2 hours and 25 minutes for this section

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:....................................

STUDENT NUMBER/NAME.............................

Section I

20 marks
Attempt Questions 1–20
Allow about 35 minutes for this section

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
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	A	B	C	D
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STUDENT NUMBER/NAME.............................

1 In an experiment to verify Galileo's statement that "all objects, regardless of their mass, will fall at the same rate", what should be controlled?

A. The masses of the objects

B. The angle at which they are dropped

C. The time for each to fall to the ground

D. The height each was dropped from

2 A ball is rolled down a slope in an experiment to determine the acceleration due to gravity.

Which of the following is most likely to make this experiment *invalid*?

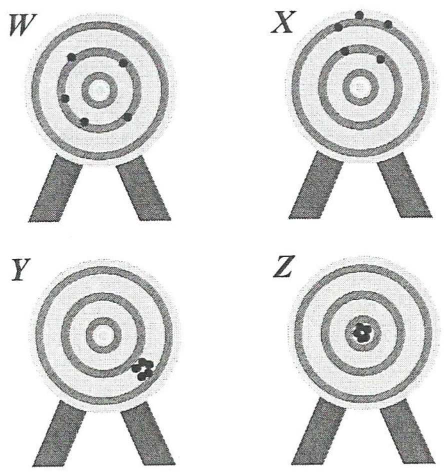
A. Using light gates to measure time instead of a stopwatch

B. Having a person in the group release the ball from the top of the slope

C. Using a very short slope length

D. Calculating the slope angle from measurements and sin relationship for angle

3 The diagrams show the results of darts being thrown at a board.



Which statement about these results is correct?

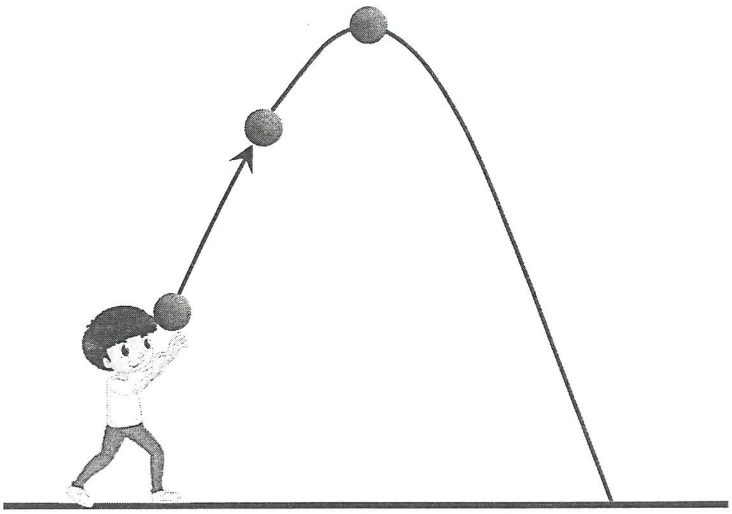
A. Results *W* are more reliable than results *Y*.

B. Results *Y* and *Z* are more reliable than *W* and *X*.

C. Results *Z* are reliable, while results *Y* are unreliable.

D. Results *X* are less reliable than results *W*.

4 The photo projectile motion.



What does the range of this projectile depend on?

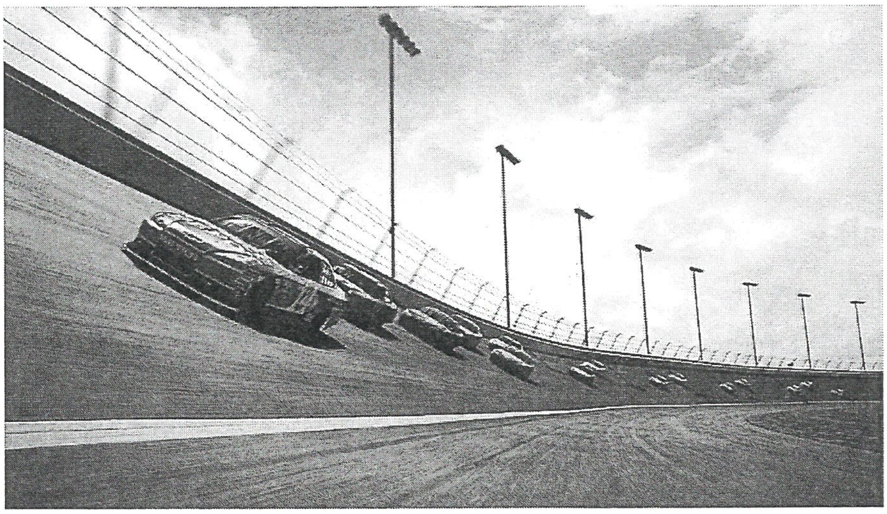
A. The initial velocity and starting height

B. The vertical component of its launch velocity and its angle of elevation

C. The horizontal component of its launch velocity and its angle of elevation

D. The maximum height above the launch position and its time of flight

5 Why are the bends on fast speed racing circuits like the one shown in the photo banked?



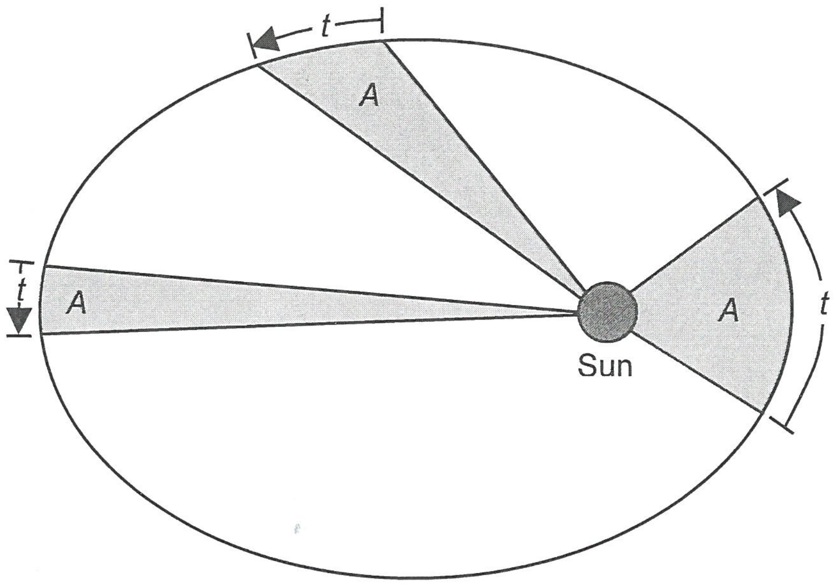
A. It increases the magnitude of the friction between the car and the surface.

B. To prevent the car turning the corner too fast.

C. Because the stability of the car depends less on the frictional force between the tyres and the surface, the tyres will last longer.

D. The horizontal component of the normal reaction force on the car provides the centripetal force.

6 What aspect of planetary motion is depicted in the diagram?



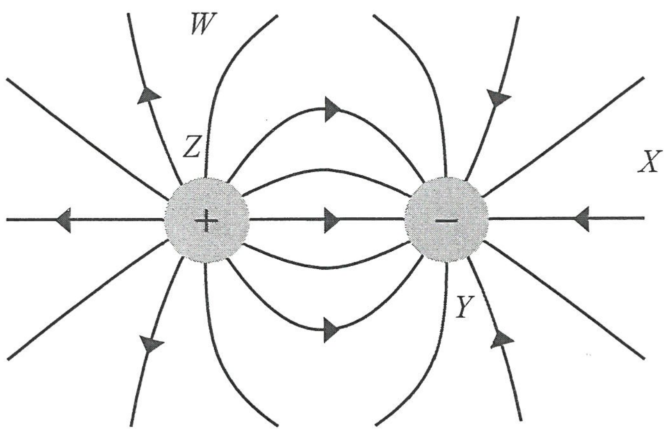
A. Kepler's first law of planetary motion

B. Kepler's second law of planetary motion

C. Kepler's third law of planetary motion

D. The faster speed of planets further from the Sun

7 Four positions are marked within the electric field shown.



A positive charge is moved from one of these positions to another position.

Which move will involve the most work being done by the field?

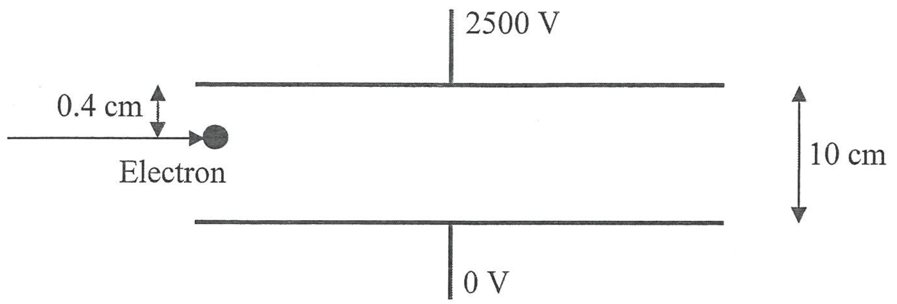
A. Moving the charge from W to Z

B. Moving the charge from Y to W

C. Moving the charge from X to Y

D. Moving the charge from Y to Z

8 An electron travels into the region between two parallel plates which are 10 cm apart and have a potential difference of 2500 V between them.



What force is on the electron when moving between the plates?

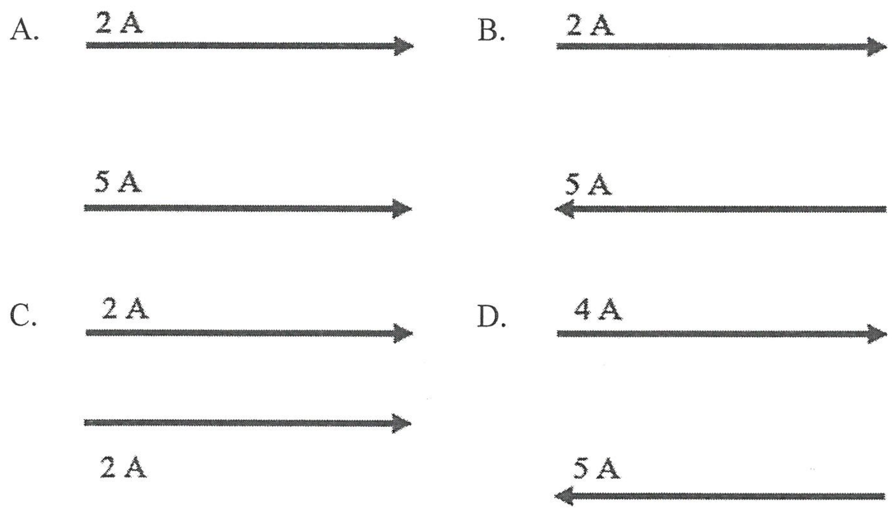
A. 4×10^{-15} N towards the bottom plate

B. 1.0×10^{-14} N towards the bottom plate

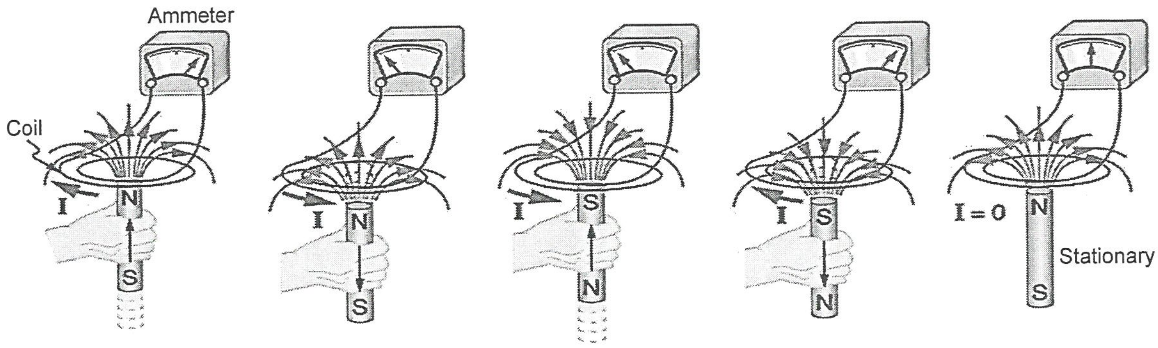
C. 4.0×10^{-15} N towards the top plate

D. 6.7×10^{-15} N towards the top plate

9 If the distance between each pair of current-carrying conductors have been drawn to scale, between which pair will the attractive force be greatest?



10 The diagram shows a bar magnet moved in or out of a wire coil attached to a galvanometer.



What is this diagram demonstrating?

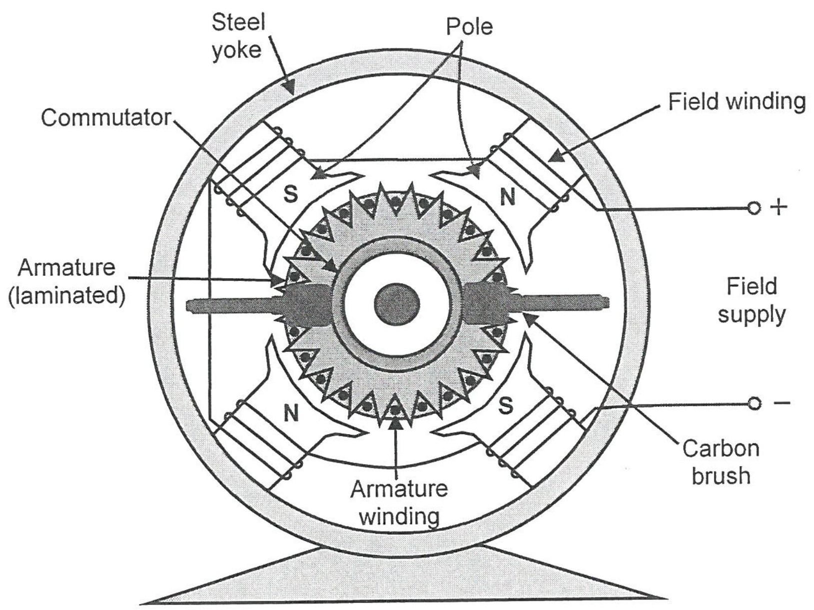
A. Electromagnetic induction

B. Ampere's law

C. Faraday's law

D. The motor effect

11 The DC motor shown has two sets of magnets.



Why are there two north opposite each other and two south poles opposite each other rather than the north and south poles opposite?

A. To increase the torque on the coil

B. To reduce the effects of the back emf

C. To ensure all the field windings are in the same direction

D. To result in torques that all cause the coil to rotate in the same direction

12 Why is it possible to identify elements using spectroscopy?

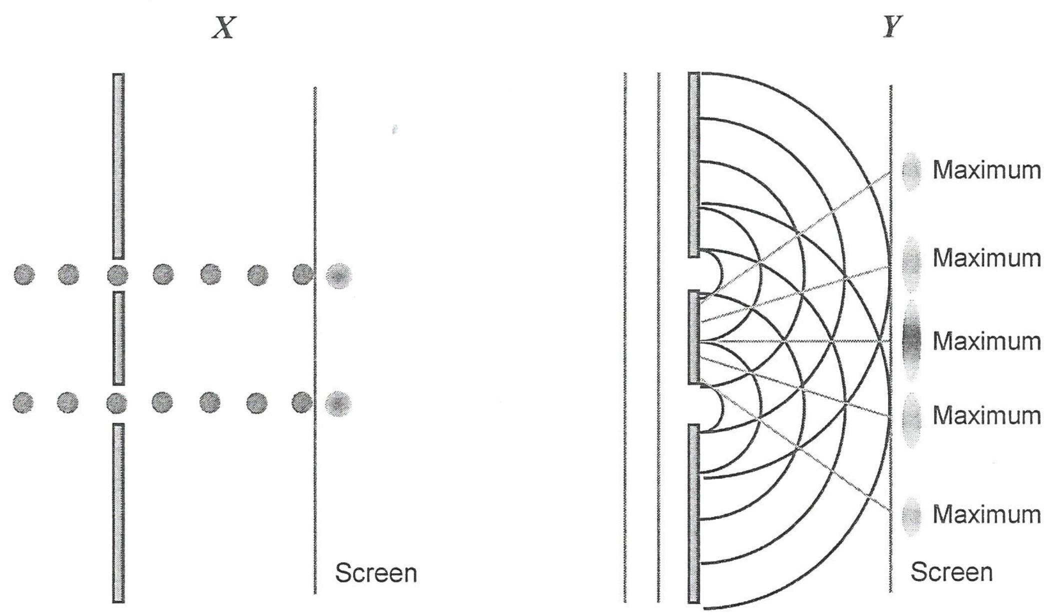
A. Every element has a unique continuous spectrum.

B. Every element has a unique set of possible energy levels for electrons.

C. Absorption spectra form dark lines on a photographic plate.

D. When electrons change energy levels, they emit energy.

13 Diagrams *X* and *Y* below depict possible results of light entering two narrow slits which are very close together.



Which statement about these diagrams is correct?

A. The particle model of light supports both diagrams.

B. The wave model of light supports both diagrams.

C. The particle nature of light is not supported by diagram *Y*.

D. Diagram *X* shows the results of Young's double slit experiment.

14 What is the key idea in Wein's law?

A. The temperature of a star is proportional to the frequency of its dominant light emission.

B. As the temperature of a hot body decreases, the wavelength of the light it emits also decreases.

C. A black body is one which absorbs all the radiation falling on it.

D. The hotter an object, the longer the wavelength of the light emitted.

15 What is an inertial frame of reference?

A. It is a frame of reference which is accelerating constantly.

B. It is a frame of reference in which Newton's laws of motion do not hold true.

C. It is a frame of reference in which inertial forces act.

D. It is a frame of reference which is at rest or moving with constant velocity.

16 Which properties of stars can be inferred from their stellar spectra?

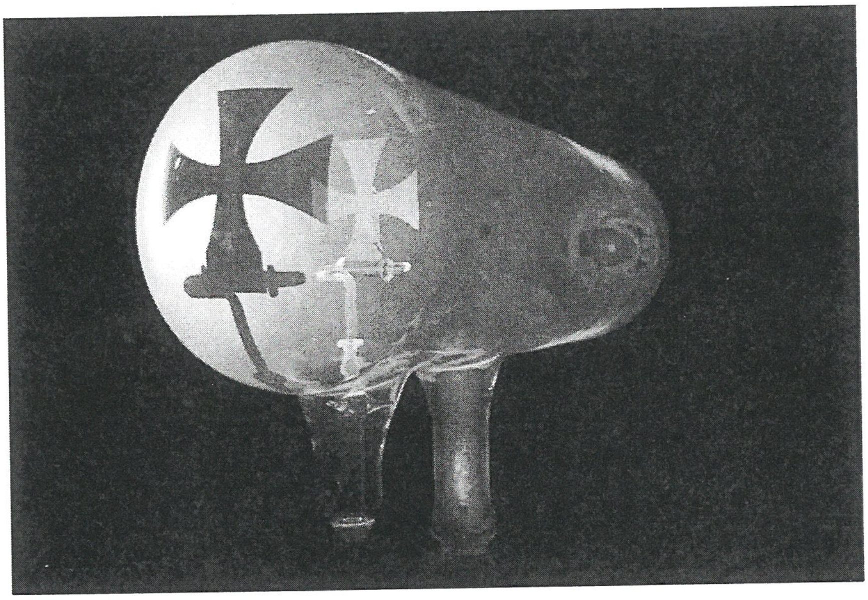
A. Temperature, size and colour

B. Core temperature and elemental composition

C. Surface temperature, elemental composition and radial speed

D. Size, motion through space and rotational velocity

17 A picture of an experiment with cathode rays is shown below.



What property of cathode rays was determined from this experiment?

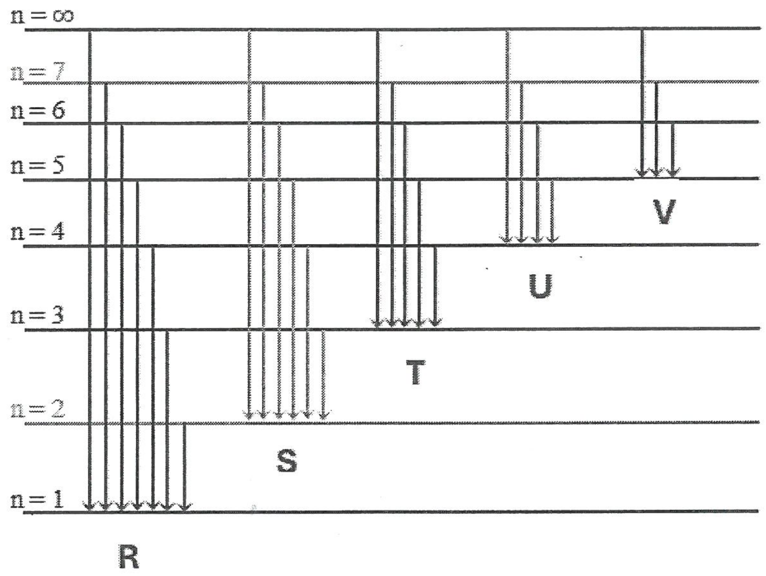
A. Cathode rays travel in straight lines.

B. Cathode rays are negatively charged.

C. Cathode rays have negligible mass.

D. Cathode rays travel from the negative to the positive electrode.

18 The electron energy level diagram for an atom is drawn below.



For this atom, which of the electron 'jumps' involves the emission of the greatest amount of energy?

A. From level 4 to level 1 in the series labelled R

B. From level 6 to level 2 in the series labelled S

C. From level 7 to level 3 in the series labelled T

D. From level ∞ to level 4 in the series labelled U

19 Which nuclear equation below is *incorrect*?

A. ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{142}_{56}Ba + {}^{91}_{36}Kr + 3{}^{1}_{0}n$

B. ${}^{234}_{90}Th \rightarrow {}^{234}_{91}Pa + {}^{0}_{-1}e$

C. ${}^{4}_{2}He + {}^{4}_{2}He \rightarrow {}^{7}_{4}Be + {}^{1}_{0}n$

D. ${}^{16}_{8}O + {}^{16}_{8}O \rightarrow {}^{28}_{14}Si + {}^{4}_{2}He + 2{}^{1}_{0}n$

20 Which of the following shows the quark composition of a neutron?

A. up – up – down

B. up – down – down

C. up – up – up

D. down – down – down

Section II

80 marks
Attempt Questions 21–34
Allow about 2 hours and 25 minutes for this section

Write your student number and/or name on every page.

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

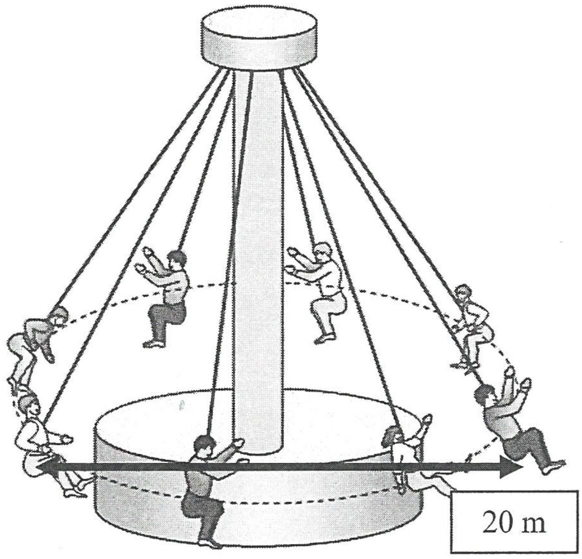
Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

STUDENT NUMBER/NAME..............................

Question 21 (6 marks)

Children are on a roundabout moving in uniform circular motion with a diameter of 20 m. Each child has a mass of 50 kg and their support rods are angled at 25° to the vertical pole at the centre of the roundabout.



(a) Calculate the centripetal force on each child. Show your working. **3**

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(b) Calculate the time it takes for each child to do one complete revolution of the roundabout. **3**

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Question 22 (4 marks)

(a) Compare classical momentum and relativistic momentum. **2**

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(b) Explain why there is a maximum speed for any mass regardless of the energy applied. **2**

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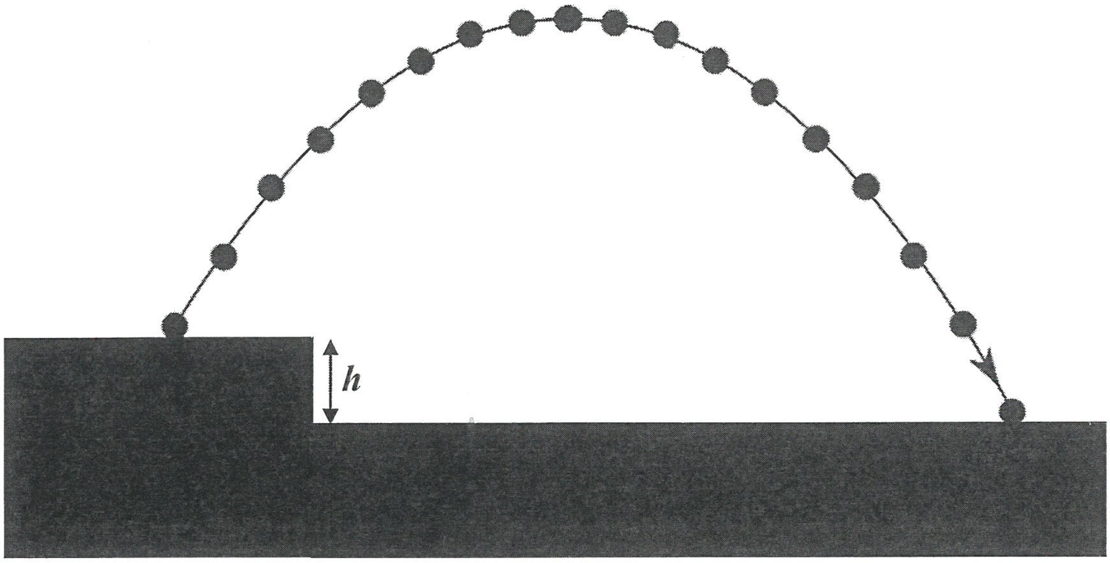
Question 23 (5 marks)

The diagram shows strobe photos of the motion of a projectile shot from a height *h*. **5**
The strobe frequency is one shot per second.



Calculate the difference in height *h*.

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Question 24 (5 marks)

Describe the orbits of satellites at a variety of distances from the Earth, including geostationary orbits. In your answer, relate satellite orbits to their uses. **5**

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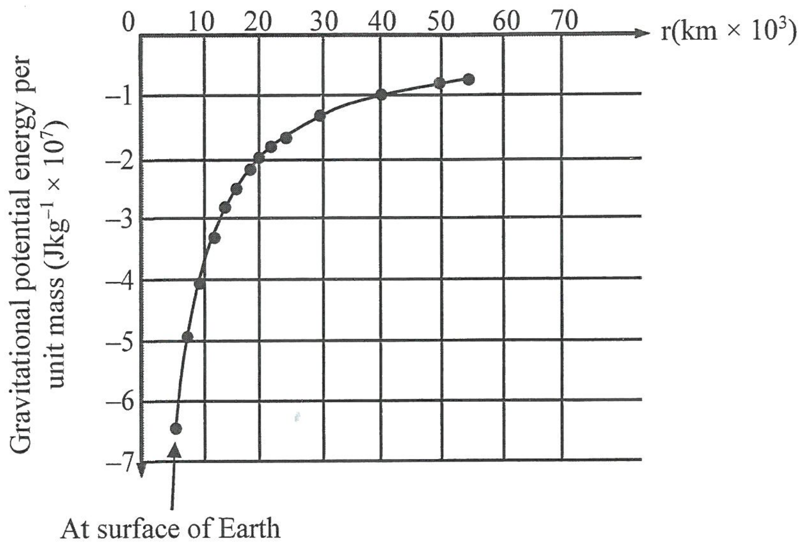
Question 25 (6 marks)

The graph shows how the gravitational potential energy of a 1 kg mass varies with its distance from the Earth's centre.



(a) Show that the gradient of a tangent at any point on this graph has the same units as the acceleration due to gravity. **3**

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(b) Determine the value of the acceleration due to gravity acting on a satellite which is in an orbit 13.6×10^3 km from the Earth's surface using an appropriate tangent drawn on the graph. **3**

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Question 26 (5 marks)

(a) What evidence led to the discovery of the expansion of the universe? **2**

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(b) How did Hubble's work expand our knowledge of the universe? **3**

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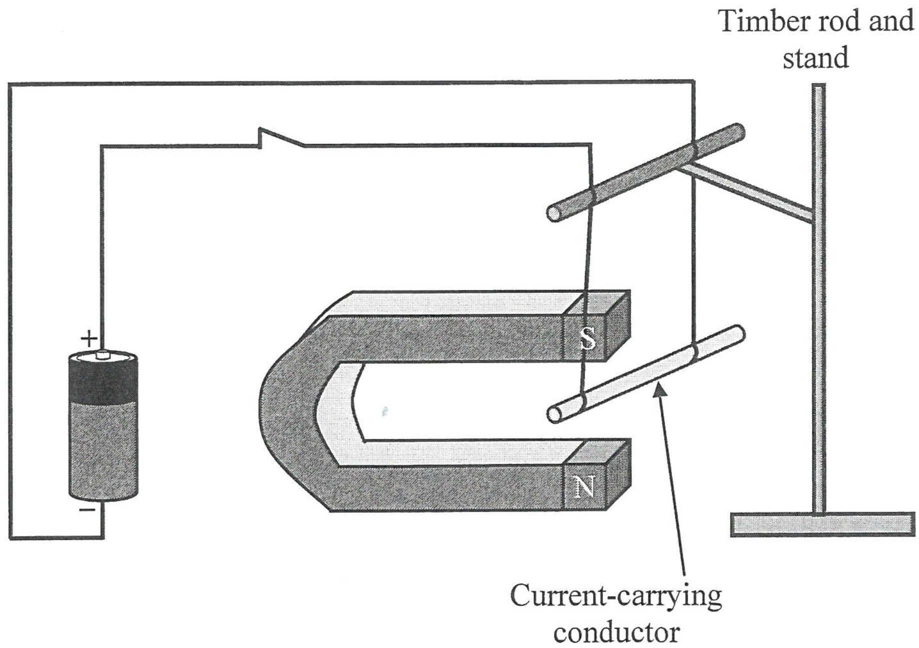
Question 27 (5 marks)

The diagram shows a conducting rod suspended by light conducting wires from a timber rod on a stand in the magnetic field produced by a horseshoe magnet. **5**



Predict and explain what happens to the rod when the tapping key is closed and a current flows in the circuit. Justify your answer.

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Question 28 (4 marks)

Discuss the production of back emf and its effects on the operation of electric motors. **4**

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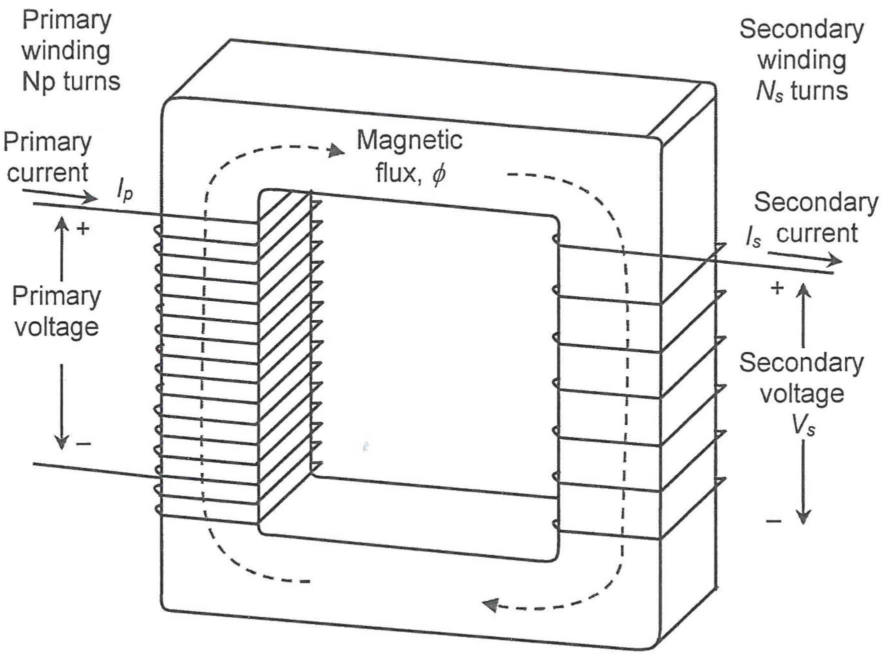
Question 29 (5 marks)

The diagram below depicts a simple transformer. **5**



The voltage relationship $\frac{V_P}{V_S} = \frac{N_P}{N_S}$ derived in textbooks only applies to an 'ideal' transformer.

Outline the shortcomings of practical transformers and the strategies used to improve their performance.

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Question 30 (7 marks)

Compare the Rutherford and Bohr models of the atom. In your answer, outline supporting evidence and limitations for each model. **7**

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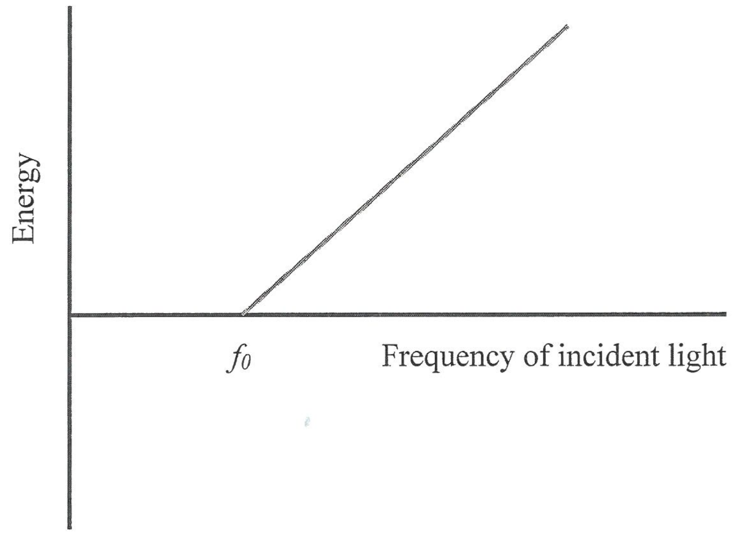
Question 31 (7 marks)

The graph relates frequency of light to maximum kinetic energy of the emitted photoelectrons from a material.



(a) On the graph, label the work function of the material and explain what is meant by the term. **2**

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(b) Outline the significance of the frequency, f_0, labelled on the graph, to our understanding of the nature of light. **5**

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Question 32 (7 marks)

Describe the method of an experiment to verify the relationship in Malus' law of polarisation. **7**

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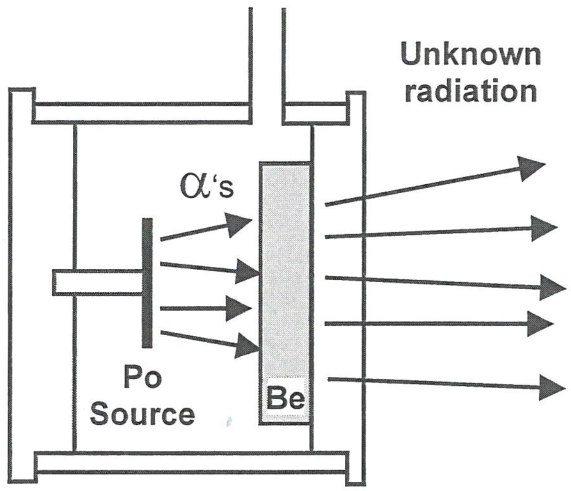
Question 33 (5 marks)

In 1930, physicists fired alpha particles at a thin foil of beryllium as depicted below and were puzzled by the neutral radiation produced. **5**



Outline how this reaction resulted in the discovery of the neutron and the involvement of conservation laws.

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Question 34 (9 marks)

Outline Maxwell's theory on the nature of light waves and the evidence supporting this theory. **9**

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End of paper

NSW INDEPENDENT TRIAL EXAMS – 2025
PHYSICS – TRIAL HSC EXAMINATION
MARKING GUIDELINES

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
D	C	B	A	D	B	C	C	A	A	D	B	C	A	D	C	A	B	D	B

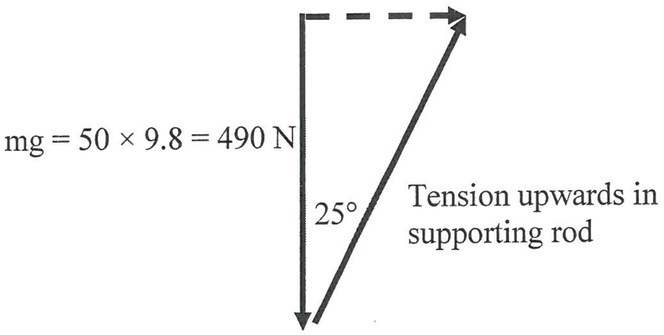
Section II

Question 21(a)

Criteria	Marks
• Calculates centripetal force by vector addition OR by analysing horizontal and vertical components	3
• Calculates centripetal force by vector addition OR by analysing horizontal and vertical components with ONE computational error OR incorrect angle	2
• Draws a triangle of vectors OR provides a component force expression	1

Sample answer:

Centripetal force: F_c = gravitational force + tension along rod



$\tan 25° = \dfrac{F_c}{mg}$ $\quad F_c = 50 \times 9.8 \times \tan 25° = 228.5$ N towards centre of roundabout

Question 21(b)

Criteria	Marks
• Calculates time for complete revolution	3
• Calculates time for complete revolution with ONE computational error OR incorrect radius	2
• Calculates orbital speed	1

Sample answer:

$F_c = ma_c$

$a_c = \dfrac{228.5}{50} = 4.57$ m s^{-2} towards the centre

and from $a_c = \dfrac{v^2}{r}$

$v = \sqrt{4.57 \times 10} = 6.76$ m s^{-1}

From $v = \dfrac{s}{t}$

Time for complete revolution $= \dfrac{2\pi r}{v}$

$= \dfrac{(2 \times 3.14 \times 10)}{6.76}$

$= 9.3$ s

Question 22(a)

Criteria	Marks
• States relative momentum allows for increased mass and explains reason for mass increase	2
• States relative momentum allows for increased mass	1

Sample answer:

Classical physics is based on Newtonian physics and assumes that the mass of a moving object remains constant so the momentum changes are proportional to the velocity changes. Relativistic momentum changes are due to both velocity increases and increases in relativistic mass of the object due to the velocity increases, as predicted in Einstein's Theory of Special Relativity. Note while the increase in energy and momentum as an object approaches the speed of light can be interpreted as an apparent increase in mass i.e. its relativistic mass, and it is more accurate to say that the object's energy and momentum are increasing, not its fundamental mass.

Question 22(b)

Criteria	Marks
• Explains speed limit reached due to need for increasingly larger amounts of energy as mass increases	2
• States due to energy needs OR increased mass	1

Sample answer:

Energy is needed to increase velocity of a mass, the amount proportional to its mass. Special relativity states that as the object's velocity increases the mass also increases requiring larger and larger amounts of energy to accelerate it until the amount required becomes infinite at the maximum speed, i.e. speed of light.

Question 23

Criteria	**Marks**
• Calculates height *h*, providing FIVE steps required	**5**
• Provides FOUR steps needed	**4**
• Provides THREE steps needed	**3**
• Provides TWO steps needed	**2**
• Provides ONE step needed	**1**

Sample answer:

Finding initial vertical velocity:

From photo, time to reach zero vertical velocity v_y at highest position = 8 s
$v_y = u_y + at$
$0 = u_y - 9.8 \times 8$
Initial vertical velocity, $u_y = 78.4$ m s^{-1} (upwards)

Finding distance to top:

$v_y^2 = u_y^2 + 2as_y$
$0 = 7.4^2 - 19.6\ s_y$
Distance up, $s_y = 313.6$ m

Finding time of fall:

From photo, time to fall from rest = 9 s

Finding distance fallen:

$s_{fall} = u_y t + \frac{1}{2}at^2$
$s_{fall} = 0 + 0.5 \times 9.8 \times 9^2$
s_{fall} distance fallen = 396.9 m

Finding height *h*:

Therefore, height $h = 396.9 - 313.6$
$= 83$ m

Question 24

Criteria	Marks
• Provides information about THREE types of orbit and detailed information of their use	4–5
• Lists THREE types of orbit and their uses	3
• Lists TWO types of orbit and TWO uses	2
• Identifies ONE type of orbit and a use	1

Sample answer:

Most artificial satellites are placed in near-circular orbits for stability. Some are in geostationary orbits at an altitude of about 35 000 km and remain above the same point on the surface of the Earth. They are used mainly for communication, TV, radio, internet services and weather monitoring and prediction, ensuring consistent altitude and coverage.

Satellites in low-earth orbits (LEO) range in altitude from about 160 to 2000 km and are used for scientific study (International Space Station, Hubble telescope). LEO satellites in polar orbits are used for spying because they can observe every part of the Earth's surface about every twelve hours.

Satellites in medium Earth orbits, between LEO and geostationary, are global positioning satellites used for navigation.

Question 25(a)

Criteria	Marks
• Provides the equation for graph gradient, derives acceleration and states units	3
• Provides the equation for graph gradient and substitutes units with ONE error	2
• Provides units for some quantities	1

Sample answer: with one error

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{y \text{ axis units}}{x \text{ axis units}}$$

$$\text{Gradient} = \frac{\text{energy} \times \text{mass}^{-1}}{\text{distance}}$$

Energy = work = force × distance

$$\text{Gradient} = \frac{\text{force} \times \text{distance} \times \text{mass}^{-1}}{\text{distance}} = \frac{\text{force}}{\text{mass}} = \text{acceleration}$$

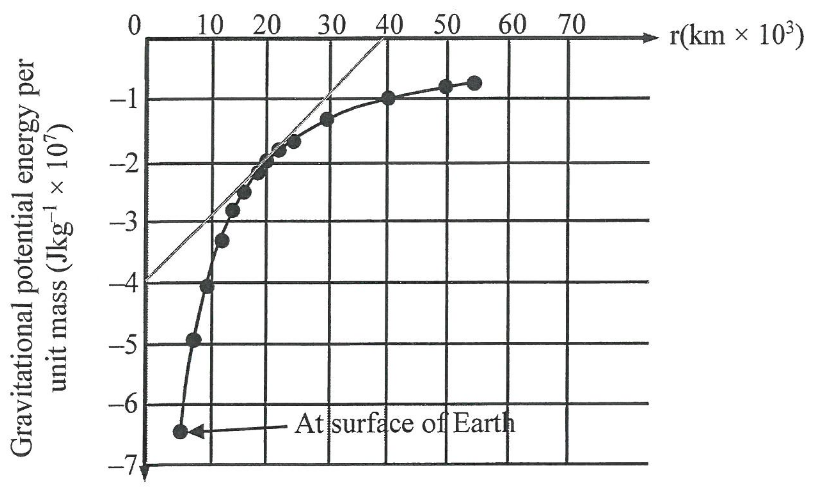
Hence, units for gradient = acceleration units.

Question 25(b)

Criteria	Marks
• Calculates acceleration from gradient of suitable tangent adding Earth radius with correct units	3
• Calculates acceleration from gradient with incorrect units OR without adding Earth radius	2
• Calculates a gradient	1

Sample answer:

Radius of Earth = 6.371×10^6 m = 6.4×10^3 km
Distance from centre of Earth = $13.6 \times 10^3 + 6.4 \times 10^3$ km = 20×10^3 km



From the graph, drawing in a tangent at radius 20×10^3 km and substituting using Δx in SI unit of metres,

$$\text{acceleration} = \frac{\Delta y}{\Delta x} = \frac{(4.0 \times 10^7)}{40 \times 10^6}$$

$$= 1.0 \text{ m s}^{-2}$$

Question 26(a)

Criteria	Marks
• Provides a clear explanation of speed and distance determination of galaxies and their relationship	2
• States evidence is relation of red shift to distance	1

Sample answer:

The key evidence that led Hubble to this discovery came from measuring the red shift in the spectral line of distant galaxies with Cepheid variable stars of known distance from Earth. The size of the red shift was used to find their velocity. Hubble found that the distance to a galaxy is directly proportional to its recessional velocity (the rate at which it is moving away) showing that the universe is expanding.

Question 26(b)

Criteria	Marks
• Explains clearly the origin of the universe from a point in space as based on the observed expansion in all directions using Hubble's law	3
• Explains clearly the origin of the universe from a point in space	2
• States work provided evidence for Big Bang theory	1

Sample answer:

Hubble's observations provided strong evidence that the universe is expanding in all directions and was once much smaller and denser, supporting the Big Bang theory of the universe's origin. Measurement of the rate of expansion (the Hubble constant) and projecting galaxies backwards at appropriate speeds provides evidence that they all started at approximately the same point in space – the position of the Big Bang some 14 billion years ago. This is estimated to be the age of the universe.

Question 27

Criteria	Marks
• Shows clear understanding of motor effect based on interaction of magnet's and conductor's magnetic fields AND correctly applies rule to predict movement outwards	5
• Provides FOUR of the following: identifies motor effect, describes magnet field of conductor, provides direction of magnet's field, interaction of fields, correctly applies rule to predict movement outwards	4
• Provides THREE of the following: identifies motor effect, describes magnet field of conductor, provides direction of magnet's field, interaction of fields, correctly applies rule to predict movement outwards	3
• Provides TWO of the following: identifies motor effect, describes magnet field of conductor, provides direction of magnet's field, interaction of fields, correctly applies rule to predict movement outwards	2
• Provides ONE of the following: identifies motor effect, describes magnet field of conductor, provides direction of magnet's field, interaction of fields, correctly applies rule to predict movement outwards	1

Sample answer:

A current carrying conductor in a magnetic field experiences a force. This is known as the motor effect. When a current flows through a conductor, a circular magnetic field is induced around that conductor. There is also magnetic field from north pole to south pole of the magnet. If the conductor is within a second magnetic field, then the two magnetic fields interact with each other to produce a force. This effect is the basis of electric motors. In the circuit provided, the current in the conductor flows from the left-hand side of the rod and the magnetic field is up from the north pole of the horseshoe magnet. Applying the right-hand palm rule, the suspended conductor will swing outwards from between the poles of the magnet.

Question 28

Criteria	Marks
• Demonstrates a clear understanding of cause of induced emf and its direction as well as the effect on the motor speed with different loads based on the law of conservation of energy	4
• Explains the induced emf and direction and the effect on the motor speed	3
• Provides TWO of the flowing: cause of induced emf, emf direction, the effect on the motor speed	2
• Provides ONE of the flowing: cause of induced emf, emf direction, the effect on the motor speed	1

Sample answer:

As the motor spins, the armature (rotor) is a conductor moving through magnetic field and therefore an emf is induced. According to Lenz's law, the direction of the emf opposes the change involved which is the applied voltage and is therefore called the back emf. This back emf increases with speed until it equals the applied emf. There is then no net emf on the rotor so the motor runs at constant speed.

The induced "back" emf in the coils must oppose the applied emf otherwise the rotor would increase in speed indefinitely and the system would not obey the law of conservation of energy. In this way, back emf provides a form of feedback control. Changes in load affect the motor speed, which in turn affects the back emf and, consequently, the current. These changes help maintain the stable operation of the motor under varying load conditions.

Question 29

Criteria	Marks
• Identifies reason for not ideal, outlines more than ONE heat energy loss, another energy loss and at least TWO strategies to reduce loss	5
• Identifies reason for not ideal, outlines TWO energy losses and ONE strategy to reduce loss OR • Identifies reason for not ideal, outlines ONE energy loss and TWO strategies to reduce loss	4
• Identifies reason for not ideal, outlines ONE energy loss and ONE strategy to reduce loss	3
• Identifies reason for not ideal, outlines ONE strategy to reduce loss	2
• Outlines ONE energy loss	1

Sample answer:

In an ideal transformer there will be no energy losses. In real transformers, some energy is lost as heat energy in the conducting wires and as eddy currents set up in the core, as well as through magnetic flux leakage from the coils.

These can be reduced but never eliminated by making the core of low resistance, high-grade silicon steel and also laminating the core which both reduce eddy current forming in the core. Energy losses in the coil windings can be reduced by using highly conducting, low resistance copper wires. Magnetic flux losses can be reduced by inter-winding the primary and secondary coils so that their insulated turns are in contact with each other, reducing the distance which the magnetic flux needs to travel. The efficiency of transformers is also improved by using an appropriate coolant to reduce heating effects within it.

Question 30

Criteria	Marks
• Provides clear details of experimental evidence for and description of EACH model, its usefulness and limitations	6–7
• Provides details of EACH model and some evidence	4–5
• Provides limited details of EACH model and evidence	2–3
• Outlines ONE model	1

Sample answer:

The Rutherford model describes atoms as having a small, dense, positively charged nucleus surrounded by electrons in orbit. The evidence for his model was based on the results of alpha particle scattering experiments, could be applied to all atoms but made no predictions.

The Bohr model retained the positively charged nucleus but had electrons orbiting in fixed orbits or energy levels. The electrons could change levels by gaining or losing energy as they rose to a higher or fell to a lower level respectively. Using Planck's quantum formula, Bohr was able to calculate the frequency of the emitted lines of the Balmer spectrum and predicted the wavelengths of other hydrogen spectra which were found, supporting his model.

The Bohr model and equations worked well for the hydrogen atom but did not work for atoms with more than one electron because it did not consider electron-electron interactions in more complex atoms.

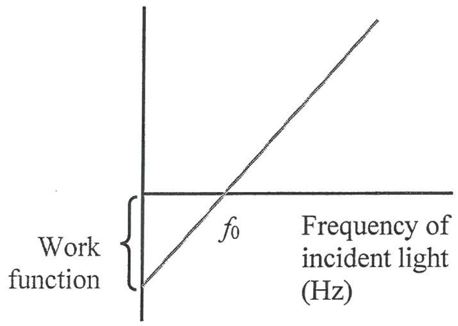
Neither model could explain the splitting of spectral lines into multiple narrower lines in the presence of a magnetic field (the Zeeman effect) or the wave-like properties of electrons which were later explained by quantum mechanics.

Question 31(a)

Criteria	Marks
• Provides definition AND correct label for work function	2
• Provides definition OR correct label for work function	1

Sample answer:

The work function is the energy needed to overcome the forces that hold an electron within a material and liberate it from the surface. The smaller the work function, the easier electrons can be emitted from the surface of the material when heat or light energy is incident on it.



Question 31(b)

Criteria	Marks
• Demonstrates a clear understanding of how quanta explains threshold frequency and resulted in the acceptance of the dual nature of light	4–5
• Uses quanta to explain the threshold frequency and resulted in the acceptance of the dual nature of light	2–3
• States that threshold frequency is evidence for particle nature of light	1

Sample answer:

f_0 is the threshold frequency of the material, equal to the minimum frequency of incident light for photoemission of electrons to occur. Einstein used Planck's hypothesis that light was quantitised, i.e. in packets to explain the photoelectric effect providing evidence of the dual nature of light. If light was simply a wave, the continuous impact of any frequency on the material surface would eventually supply sufficient energy for the emission of electrons, meaning that electrons would be emitted at all frequencies. This does not happen. Light having a particle nature as well as a wave nature also explained the puzzle of black body radiation.

Question 32

Criteria	Marks
• Provides a clear outline of the steps required, including the equipment, the independent variable, tabulating of results, graphing of results, gradient determination and necessary result to verify law	**6–7**
• Provides an outline of the steps required with ONE of the following missing: the equipment, the independent variable, tabulating of results, graphing of results, gradient determination and necessary result to verify law	**5**
• Provides an outline of the steps required with TWO of the following missing: the equipment, the independent variable, tabulating of results, graphing of results, gradient determination and necessary result to verify law	**4**
• Provides an outline of the steps required with THREE of the following missing: the equipment, the independent variable, tabulating of results, graphing of results, gradient determination and necessary result to verify law	**3**
• Provides several details for experiment	**1–2**

Sample answer:

Malus' law states that $I = I_{max}\cos^2\theta$ where θ is the angle between planes of polarisation of two light filters.

The experiment requires an unpolarised light source, two polarising light filters and a light meter set up in a dark room.

Both polarising filters are placed in front of the light source with their planes of polarisation parallel, that is, with angle θ equal to zero. The intensity of the transmitted light recorded from the light meter is the maximum light transmission, I_{max}.

The second polarising filter is rotated 15, 30, 45, 60, 75 and 90 degrees and the intensity of light transmission recorded at each angle θ in a table.

The intensity of light transmission I is plotted against the square of the cosine of the angles of rotation. If the graphed points justify a straight line of best fit through the origin, then the transmitted intensity I is proportional to the square of the cosine of the angle. The gradient of the line should also agree with the measured maximum light transmission, I_{max} for results to verify Malus' law.

Question 33

Criteria	Marks
• Provides a detailed sequence leading to the identification of the neutron including the properties of the radiation, bombarding paraffin, resulting dilemma, Chadwick's work and Rutherford's prediction	**5**
• Provides details of FOUR of the following: properties of radiation, bombarding paraffin, resulting dilemma, Chadwick's work, Rutherford's prediction, reaction equation	**4**
• Provides details of THREE of the following: properties of radiation, bombarding paraffin, resulting dilemma, Chadwick's work, Rutherford's prediction, reaction equation	**3**
• Provides details of TWO of the following: properties of radiation, bombarding paraffin, resulting dilemma, Chadwick's work, Rutherford's prediction, reaction equation	**2**
• Provides details of ONE of the following: properties of radiation, bombarding paraffin, resulting dilemma, Chadwick's work, Rutherford's prediction, reaction equation	**1**

Sample answer:

The radiation had greater penetrating power than gamma rays and was not a charged particle. Other physicists projected the radiation onto paraffin resulting in the emission of protons. The energy of the protons was calculated using conservation of energy and momentum laws. This determined such a high energy for the radiation from beryllium that either a gamma ray nature or conservation laws had to be ruled out.

Chadwick repeated the experiments and carried out other interaction experiments. He estimated the mass of the "radiation" from collisions with nuclei of different masses, again using conservation laws, and concluded that the beryllium emission was a neutral particle with a mass approximately equal to that of the proton. Chadwick realised that the radiation was the neutral particle which had been predicted by Rutherford and called the neutron.

The equation for the reaction producing the neutron was realised to be:

$$^{9}_{4}\text{Be} + ^{4}_{2}\text{He} \rightarrow ^{12}_{6}\text{C} + ^{1}_{0}\text{n}$$

Question 34

Criteria	Marks
• Provides a clear description of Maxwell's theory, Maxwell's calculation of the speed of light and prediction of other electromagnetic waves	8–9
• Provides basic details of Maxwell's theory and identifies supporting evidence	6–7
• Provides name of waves with minimal detail of formation and some supporting evidence	4–5
• Provides name of waves with minimal detail of formation	3
• Provides name of waves and relates waves to light	2
• Provides name of waves	1

Sample answer:

Maxwell formulated a mathematical theory of electromagnetic waves which unified the theories of electricity, magnetism and light as well. Using four equations now known as Maxwell's Equations, he described and quantified the relationships between these three areas of physics. His equations predicted the propagation of electromagnetic energy as waves.

He proposed that a changing magnetic field would induce a changing electric field which, in turn, induced a changing magnetic field and so on. The fields vibrated perpendicular to one another and to the direction of propagation of the wave itself in the form of an electromagnetic transverse wave which self-propagated.

Maxwell's wave equations showed that the speed of the waves was a combination of constants in electricity and magnetism. The calculated speed of his electromagnetic waves was about 3×10^8 metres per second which agreed with the known speed of light, providing evidence for his theory.

Using his theory, Maxwell predicted that there would be a continuous frequency range of electromagnetic radiation extending beyond the visible light spectrum at both ends. Such waves were found and provided further evidence for the theory.

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Question	Marks	Content module		Syllabus Outcomes (PH)	Targeted performance bands
Section I					
1	1	5	Gravitational Field	11/12-2, 12-12	3-4
2	1	5	Gravitational Field	11/12-2, 12-12	3-4
3	1	5	Gravitational Field	11/12-2, 12-12	2-3
4	1	5	Projectile Field	12-12	4-5
5	1	6	Circular Motion	12-12	4-5
6	1	6	Circular Motion	11/12-4, 12-13	3-4
7	1	6	Charged Particles, Conductors and Electric and Magnetic Fields	11/12-6, 12-13	4-5
8	1	6	Charged Particles, Conductors and Electric and Magnetic Fields	11/12-6, 12-13	2-3
9	1	6	The Motor Effect	11/12-6, 12-13	2-3
10	1	6	Electromagnetic Induction	11/12-6, 12-13	3-4
11	1	7	Applications of the Motor Effect	11/12-7, 12-13	3-4
12	1	7	Electromagnetic Spectrum	11/12-7, 12-14	2-3
13	1	7	Light: the Wave Model	11/12-7, 12-14	4-5
14	1	7	Light: the Quantum Model	11/12-7, 12-14	3-4
15	1	7	Light and Special Relativity	11/12-7, 12-14	5-6
16	1	8	Origin of the Elements	11/12-7, 12-15	4-5
17	1	7	Structure of the Atom	11/12-6, 12-15	3-4
18	1	8	Quantum mechanical nature of the Atom	11/12-7, 12-15	2-3
19	1	8	Properties of the Nucleus	11/12-6, 12-15	3-4
20	1	8	Deep Inside the Atom	11/12-7, 12-15	3-4

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MAPPING GRID cont'd

Question	Marks	Content module		Syllabus Outcomes (PH)	Targeted performance bands
Section II					
21(a)	3	5	Circular Motion	11/12-6, 12-12	4-5
21(b)	3	5	Circular Motion	11/12-6, 12-12	3-4
22(a)	2	7	Light and Special Relativity	11/12-7, 12-15	3-4
22(b)	2	7	Light and Special Relativity	11/12-7, 12-15	4-5
23	5	5	Projectile Motion	11/12-6, 12-12	4-5
24	5	5	Motion in a Gravitational Field	11/12-7, 12-12	4-5
25(a)	3	5	Motion in a Gravitational Field	11/12-5, 12-12	4-5
25(b)	3	5	Motion in a Gravitational Field	11/12-5, 12-12	4-6
26(a)	2	8	Origin of Elements	11/12-7, 12-15	3-4
26(b)	3	8	Origin of Elements	11/12-7, 12-15	3-5
27	5	6	The Motor Effect	11/12-7, 12-13	3-4
28	4	6	Applications of the Motor Effect	11/12-6, 12-15	3-5
29	5	6	Electromagnetic Induction	11/12-7, 12-13	3-5
30	7	8	Quantum Mechanical Nature of the Atom	11/12-7, 12-15	2-5
31(a)	2	7	Light: Quantum Model	11/12-5, 12-14	3-4
31(b)	5	7	Light: Quantum Model	11/12-7, 12-14	2-5
32	7	7	Light: Wave Model	11/12-2, 11/12-3, 12-14	2-5
33	5	7	Structure of the Atom	11/12-7, 12-15	2-4
34	9	7	Light: Wave Model	11/12-7, 12-14	2-5